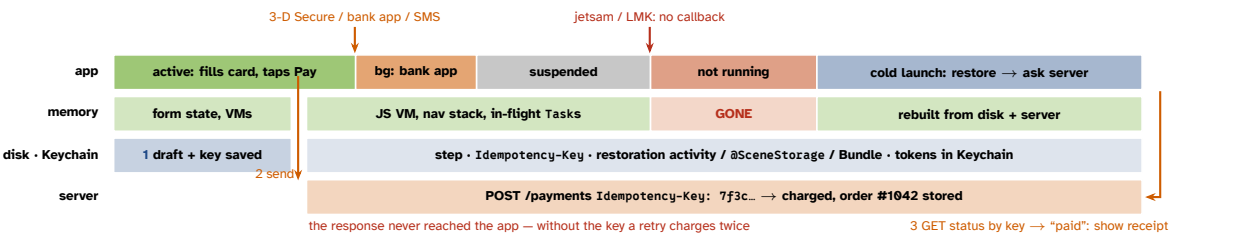


**In one line:** A backgrounded app is **suspended**, and a suspended app can be **killed without any callback** (iOS jetsam, Android low-memory killer). When the user comes back it is a **cold launch**: everything in **memory is gone**, everything on **disk / Keychain** / **restoration data survives** — and the **server may already have done the payment**. So persist the screen *and* the draft before leaving, and make the operation **idempotent** so resuming can ask “did it happen?” instead of doing it twice.



Ways an app dies

| kill                        | callback?                                            | restore?                          |
|-----------------------------|------------------------------------------------------|-----------------------------------|
| iOS jetsam (suspended)      | none; applicationWillTerminate only if running in bg | yes: restoration data kept        |
| iOS watchdog / crash        | none                                                 | crash in restore → data discarded |
| user swipes app away        | none reliable                                        | <b>no</b> — user meant “quit”     |
| Android process death (LMK) | none; onSaveInstanceState ran at onStop              | yes: task + Bundles               |
| Android config change       | onDestroy + recreate                                 | VM survives (not death)           |
| reboot / OS / app update    | none                                                 | disk only                         |

- **iOS:** the system relaunches into the saved scene; you rebuild it from stateRestorationActivity / @SceneStorage (small UI state: ids, step, tab) + the draft on disk. Save on sceneDidEnterBackground, not in willTerminate.
- **Android:** the task and back stack come back; the *top* Activity is recreated from its Bundle and a **new** process — Application.onCreate runs again, every singleton and static is fresh. SavedStateHandle / rememberSaveable hold ids; big data goes to disk (Bundle ≈ 1 MB Binder limit).
- **React Native:** the JS VM is gone; the app boots JS from scratch at the initial route. Persist navigation state (React Navigation initialState + onChange) and drafts yourself (AsyncStorage / MMKV; secrets in Keychain/Keystore). On Android, **react-native-screens** discards the restored fragment state (RNScreensFragmentFactory) to avoid crashes — so native restore will not bring your screens back.

What survives

| survives                                | lost                                    |
|-----------------------------------------|-----------------------------------------|
| files, DB, UserDefaults already written | objects, caches, singletons, JS state   |
| Keychain / Keystore items               | in-flight requests on a default session |
| restoration data, Bundles (system kill) | timers, Tasks, coroutines, listeners    |
| background URLSession transfers         | “we’ll save it in willTerminate”        |
| <b>the server’s side of the request</b> | the response you never received         |

Test it — on purpose

- **iOS:** background the app, then Stop in Xcode (or xcrun simctl terminate booted com.example.app), relaunch. Simulator *Debug* → *Simulate Memory Warning* tests *warnings*, not death. Swipe-away is *not* the test.
- **Android:** adb shell am kill com.example.app while in the background = real process death; Developer options *Don’t keep activities* only destroys Activities; *Background process limit: none* for soak testing.
- Automate one UI test: fill the form, kill, relaunch, assert the draft.

Design checklist — a death-proof flow

1. Every screen can be rebuilt from **route + ids** — no object handed over in memory that exists nowhere else.
2. Drafts persisted on change (debounced), *before* leaving the app.
3. Side-effecting calls carry an idempotency key; the server has a **status endpoint** for it.
4. Session state re-hydrates lazily (token from Keychain, user from cache) — never “set by the login screen”.
5. Restored ids validated; missing → a sane fallback screen.
6. Drafts cleared on success, cancel and logout (privacy).

Example — a checkout that survives

```
struct CheckoutDraft: Codable {
  var cartID: String, step: Int
  var payKey: UUID? // set once, at "Pay"
}

@MainActor final class Checkout {
  var draft: CheckoutDraft { didSet { store.save(draft) } }
  func pay() async throws { // key saved BEFORE the call
    let key = draft.payKey ?? UUID(); draft.payKey = key
    try await api.pay(draft.cartID, idempotencyKey: key)
  }
  func resume() async throws { // on cold launch
    guard let key = draft.payKey else { return }
    switch try await api.paymentStatus(key) {
    case .paid: route(.receipt); store.clear()
    case .notFound, .failed: route(.pay) // safe to retry
    case .pending: route(.waiting) }
  }
}

<NavigationContainer // React Navigation
  initialState={deepLink ? undefined : saved}
  onChange={s => storage.set("nav", JSON.stringify(s))}>
```

Payments and purchases

- **Idempotency key:** client makes one UUID per *intent*, persists it with the draft, sends it on every retry (Idempotency-Key on POST/PATCH); the server stores the result per key and replays it. On resume: **query status**, never blindly re-POST.
- Server truth wins: order state comes from the server / webhook, not from “the app saw 200”. Redirect flows (3-D Secure, bank app) return via a universal link into a possibly *cold* app — the route must work from zero.
- **StoreKit 2:** a purchase interrupted by a kill is re-delivered through Transaction.updates (listen from launch) until you finish() it — grant content first, then finish.

Traps

- ! Relying on applicationWillTerminate / onDestroy — not called on a system kill of a suspended app.
- ! Restoring into a screen whose data is gone — validate ids, fall back.
- ! Relaunched in the background before first unlock: .complete files and WhenUnlocked Keychain items are unreadable.
- ! Persisting RN nav state that crashes on restore — the user is stuck; reset it from an error boundary.
- ! Android: a singleton filled by the splash/login screen is null when the process is restored straight onto screen 3 — classic NPE.

Remember

**Memory dies, disk lives, the server remembers.** Save before you send; ask before you retry.